

**BUDDY: An Intelligent Robot Dog for Offline Voice
Interaction, Visual Perception, and Safety-Aware
Multimodal Control**

INDIVIDUAL THESIS

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Individual component: Autonomous Navigation, Obstacle Avoidance,
Human Following, and 2D Path Planning for an Intelligent Companion
Robot

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

This thesis presents the individual research and engineering contribution to BUDDY concerned with autonomous navigation, obstacle awareness, human following, patrol-oriented behaviour, and safety-aware movement. The work addresses the challenge of enabling a low-cost companion robot to move through indoor environments while responding to dynamic obstacles and maintaining a clear separation between high-level navigation intent and physical motor authority. The implemented research combines four-direction ultrasonic sensing, ROS 2 movement supervision, simulated grid-based path planning, collision-aware navigation, and a follow-me behaviour model. A 3D-visualised 2D navigation simulation was developed in Python using Ursina to evaluate navigation logic before complete physical deployment. The simulation incorporates A* path planning, obstacle-clearance cost weighting, field-of-view and occlusion checks, PID-based following distance, target-loss recovery, search and waiting states, and dynamic replanning. Physical prototype experiments verified four-direction ultrasonic sensing and demonstrated that active forward movement can be interrupted when an obstacle is introduced and resumed after obstacle removal through the independent safety supervisor. The broader design also proposes a lightweight 2D occupancy-style planning method that combines ultrasonic range observations with Raspberry Pi Camera 2 visual cues; this is presented as continuing work rather than a fully validated mapping result. The component contributes a modular navigation architecture suitable for further patrol, localization, mapping, and security-event behaviour on BUDDY.

Keywords: autonomous navigation, obstacle avoidance, human following, A* path planning, ROS 2

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TABLE OF CONTENTS

1 INTRODUCTION	1
1.1 Overall project context.....	1
1.2 Individual component context.....	1
1.3 Problem statement.....	2
1.4 Research gap	2
1.5 Aim and objectives.....	3
1.6 Research questions or hypotheses	3
1.7 Scope limitations and assumptions	4
1.8 Individual contribution.....	4
1.9 Thesis structure	4
2 LITERATURE REVIEW AND RELATED WORK	5
2.1 Review method and scope.....	5
2.2 Domain and theoretical foundations	5
2.3 Related research	6
2.4 Existing systems products and techniques	6
2.5 Gap synthesis and conceptual framework.....	7
3 RESEARCH METHODOLOGY	8
3.1 Research design.....	8
3.2 Data participants or study materials	8
3.3 Experimental or study procedure	8
3.4 Measures and evaluation metrics	9
3.5 Data analysis	9
3.6 Ethics privacy security and responsible AI.....	10
3.7 Reproducibility and research artefacts	10

4 PRODUCT AND SOLUTION DESIGN.....	11
4.1 Requirements.....	11
4.2 Component architecture within the complete solution.....	11
4.3 Design alternatives and decisions	12
4.4 Detailed design.....	13
4.5 Standards and professional practice	14
5 IMPLEMENTATION INTEGRATION AND TESTING.....	15
5.1 Development environment and technology stack	15
5.2 Component implementation	15
5.3 Integration with the other components.....	16
5.4 Verification and software testing	17
5.5 Deployment and operation	18
6 RESULTS AND VALIDATION	19
6.1 Research results.....	19
6.2 Product and technical results.....	19
6.3 User or stakeholder validation	20
6.4 Integration results.....	20
6.5 Summary of findings.....	20
7 DISCUSSION	21
7.1 Interpretation of findings.....	21
7.2 Comparison with literature baselines and existing solutions	21
7.3 Research contribution.....	21
7.4 Product contribution.....	22
7.5 Limitations and threats to validity.....	22
7.6 Implications.....	23
8 COMMERCIALISATION AND SUSTAINABILITY	24

8.1 Value proposition and target users	24
8.2 Market and competitive position.....	24
8.3 Business or sustainability model.....	24
8.4 Scalability, intellectual property and compliance	24
8.5 Contribution to the Sustainable Development Goals	25
9 CONCLUSION AND FUTURE DIRECTIONS	26
9.1 Conclusion	26
9.2 Achievement of objectives	26
9.3 Individual research and product contributions	26
9.4 Future directions.....	27
REFERENCES.....	28
GLOSSARY.....	30
APPENDICES	31
Appendix A Research instruments or extended results.....	31
Appendix B Academic integrity and AI use disclosure	32

LIST OF FIGURES

Figure 4.1 Navigation and safety architecture for the individual component.....	12
Figure 4.2 Follow-me behaviour state machine implemented in simulation.....	13
Figure 9.1 Proposed low-cost 2D occupancy-style planning concept for continuing work.....	27

LIST OF TABLES

Table 3.1 Evidence classification used in this thesis.....	9
Table 4.1 Navigation component requirements.....	11
Table 5.1 Development and implementation stack.....	15
Table 5.2 Navigation verification summary.....	18
Table 6.1 Summary of validated and implemented navigation results.....	20

LIST OF ABBREVIATIONS

Abbreviation	Meaning
A*	A-star heuristic path-planning algorithm
FOV	Field of View
HC-SR04	Ultrasonic distance sensor used for directional obstacle sensing
IMU	Inertial Measurement Unit
LiDAR	Light Detection and Ranging
PID	Proportional-Integral-Derivative control
ROS 2	Robot Operating System 2
SLAM	Simultaneous Localization and Mapping

1 INTRODUCTION

1.1 Overall project context

BUDDY is an intelligent robot-dog research prototype designed to combine natural interaction, visual awareness, environmental sensing, and physical mobility on affordable embedded hardware. The complete project is implemented around a Raspberry Pi 4 and ROS 2 and is organised into three principal research streams: offline auditory interaction, visual and facial intelligence, and environment-aware navigation and physical control [1]. The complete system is intentionally modular so that individual perception and intelligence components can be developed independently while still communicating through controlled interfaces.

For an embodied companion robot, movement is not simply a user-interface output. A physical platform must interpret environmental conditions continuously, avoid collision, respond to sensor uncertainty, and preserve safe behaviour even when a high-level command remains active. BUDDY therefore separates high-level requests from final motor authority. Movement requests are evaluated by an independent safety supervisor before reaching the motor executor, and active movement is reevaluated when new obstacle information arrives [1].

This individual thesis focuses on the navigation and environmental-intelligence stream. The contribution covers obstacle-aware movement, safe stopping, navigation-state design, human-following behaviour, A* route planning in simulation, dynamic replanning, and the continuing development of low-cost 2D indoor planning using ultrasonic and camera observations.

1.2 Individual component context

The individual component was initially defined as Autonomous Navigation and Obstacle Avoidance. During development, its practical scope expanded to include movement-state logic, follow-me behaviour, patrol-oriented research, path planning, recovery when a target is lost, and integration requirements for security-oriented movement. The component also includes product-level contribution to the BUDDY project website and navigation/security requirements for the companion application.

The navigation stream is intentionally divided between physically validated functions and continuing research. Four-direction ultrasonic sensing and dynamic obstacle intervention have been demonstrated on the physical robot. A more advanced 3D-visualised 2D navigation simulation has been implemented to investigate A* planning, obstacle clearance, field-of-view constraints, human following, search behaviour, and dynamic replanning. Full autonomous localization and complete patrol execution on the physical robot remain continuing work, consistent with the integration boundaries reported in the group research paper [1].

1.3 Problem statement

Low-cost companion robots require autonomous movement capabilities that are sufficiently responsive for indoor operation but computationally practical on embedded hardware. The challenge is complicated by dynamic obstacles, narrow indoor spaces, uncertain range measurements, temporary loss of a human target, and the safety risk created when high-level behaviour is coupled directly to physical motors. A navigation solution must therefore do more than generate a path: it must combine perception, route planning, reactive avoidance, target following, safe stopping, and recovery behaviour while preserving a safety boundary between navigation intent and actuation.

1.4 Research gap

Established mobile-robot research provides strong foundations for occupancy grids, graph search, local obstacle avoidance, probabilistic localization, and modern ROS 2 navigation. However, many complete navigation stacks assume richer sensing such as LiDAR, depth cameras, wheel odometry, or well-calibrated localization. BUDDY targets a lower-cost companion platform using ultrasonic range sensors and a monocular camera, with computation constrained to a Raspberry Pi-class device. The research gap addressed by this component is therefore the practical integration of lightweight planning and human-following logic with a low-cost sensing configuration and an explicit safety-supervision layer.

1.5 Aim and objectives

The aim of this research is to design and evaluate a low-cost, safety-aware navigation component that enables BUDDY to move through indoor environments, respond to obstacles, follow a human target, and provide a foundation for patrol and 2D route-planning behaviour.

- Design a navigation architecture that separates high-level movement intention from final physical motor authorization.
- Integrate four directional ultrasonic sensors for obstacle awareness and safety intervention.
- Develop and evaluate a 3D-visualised 2D navigation simulation for A* path planning, collision avoidance, and human-following behaviour.
- Implement a follow-me state machine that handles target acquisition, target loss, search, signaling, waiting, and reacquisition.
- Use PID-based distance regulation and dynamic replanning to achieve smoother following behaviour in simulation.
- Investigate a low-cost 2D occupancy-style planning approach using ultrasonic range observations and Raspberry Pi Camera 2 visual cues.
- Define integration interfaces for patrol, remote control, security events, and the wider BUDDY ROS 2 system.

1.6 Research questions or hypotheses

RQ1: How can obstacle-aware autonomous movement be implemented on a low-cost companion robot while keeping high-level navigation logic separated from final motor authority?

RQ2: Can a lightweight grid-based A* planner, combined with local obstacle avoidance and PID-based following, provide useful indoor navigation behaviour in a simulated companion-robot environment?

RQ3: What low-cost sensor-fusion architecture could support future 2D local planning using ultrasonic range measurements and monocular camera cues?

1.7 Scope limitations and assumptions

The scope is limited to indoor companion-robot navigation. The physical prototype currently validates directional obstacle sensing and safety-mediated movement intervention. Complete SLAM, metric localization, long-duration autonomous patrol, and statistically characterised navigation accuracy are outside the validated scope of this thesis version. The 3D simulation is used to evaluate algorithmic behaviour but is not treated as equivalent to physical-world validation. The proposed camera-plus-ultrasonic 2D planning approach is continuing work and is not presented as a validated 3D reconstruction system.

1.8 Individual contribution

The candidate designed and implemented the navigation simulation and contributed the navigation/security research direction of BUDDY. The primary individual artefact is `buddy_follow_me.py`, a large Ursina-based 3D simulation implementing indoor geometry, A* route planning, obstacle-clearance cost weighting, PID-based following distance, field-of-view checks, occlusion reasoning, dynamic replanning, and a multi-state target-following behaviour. The candidate also contributed navigation and security requirements to the BUDDY product website and companion-control concept. Physical ultrasonic sensing and safety-mediated movement were integrated into the shared robot platform with support from the research team.

1.9 Thesis structure

Chapter 2 reviews navigation, mapping, human-following, and ROS 2 foundations. Chapter 3 defines the research methodology and evaluation boundaries. Chapter 4 presents the component and solution design. Chapter 5 documents implementation, integration, and testing. Chapter 6 reports physically validated and software-level results. Chapter 7 discusses findings, contributions, and limitations. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the work and defines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review focuses on five areas directly related to the component: mobile-robot navigation, graph-based path planning, occupancy-grid mapping, reactive obstacle avoidance, and ROS 2 navigation architectures. Priority was given to foundational peer-reviewed work and current official ROS 2/Nav2 documentation. The review also considers companion-robot requirements because navigation quality must be interpreted in the context of safe human-centred operation rather than only shortest-path performance.

2.2 Domain and theoretical foundations

Autonomous mobile navigation typically combines perception, localization, mapping, planning, and control. Probabilistic robotics treats uncertainty as a central property of real-world sensing and action, and occupancy grids provide a common representation in which space is divided into cells associated with occupied, free, or unknown states [2], [3]. The early sonar mapping work of Moravec and Elfes demonstrated how repeated range measurements can be integrated into rasterized environmental maps, providing an important conceptual basis for low-cost range-sensor mapping [2].

A* remains one of the most influential heuristic graph-search methods. Hart, Nilsson, and Raphael formalised the use of heuristic information for minimum-cost path search [4]. In a grid map, A* can be used to search traversable cells while a cost model penalises cells near obstacles. This is directly reflected in the simulation developed in this research, where occupancy and clearance information influence route selection rather than treating every free cell as equally desirable.

Reactive obstacle avoidance complements global planning by responding to immediate environmental changes. The Vector Field Histogram and Dynamic Window Approach are influential examples of methods that combine goal-directed motion with collision avoidance [5], [6]. BUDDY does not reproduce either algorithm directly; instead, its simulation adds local repulsive obstacle influence to

planned direction, while the physical platform uses an independent safety supervisor to interrupt unsafe movement.

2.3 Related research

Social and companion robots operate in shared spaces where navigation cannot be separated from human interaction. Socially assistive robotics literature emphasises autonomy, embodiment, sensing, and actuation as connected system properties [7], [8]. Person-following research further shows that target tracking, sensor choice, occlusion handling, following geometry, and recovery behaviour are recurring design challenges [4]. Human-aware navigation research also highlights that navigation around people must consider predictability, comfort, and social constraints rather than shortest-path efficiency alone [2]. For BUDDY, this means following a person is not simply a geometric tracking problem. The system must tolerate occlusion, maintain an appropriate following distance, avoid furniture, search when the target is lost, and eventually support safe patrol behaviour.

ROS 2 provides modular communication between independently developed robotic components [11]. Modern Nav2 builds on this approach with lifecycle-managed servers and a structured navigation architecture [12]. BUDDY adopts the same general philosophy of modular perception, decision, safety, and actuation separation, although the present research uses a custom lightweight implementation rather than claiming complete Nav2-based autonomous navigation.

2.4 Existing systems products and techniques

Commercial mobile robots often rely on LiDAR, stereo or RGB-D cameras, wheel odometry, and dedicated mapping hardware. These sensors simplify metric mapping and localization but increase cost, integration complexity, and power demand. BUDDY instead investigates whether a useful companion-robot navigation layer can be progressively developed from four HC-SR04 ultrasonic sensors, a Raspberry Pi Camera Module 2, and embedded processing. The Camera Module 2 uses the Sony IMX219 image sensor and provides an 8-megapixel monocular visual stream [13]. Because it is monocular, it does not directly output metric depth; any spatial

inference from images must therefore be treated as visual estimation rather than raw 3D ranging.

2.5 Gap synthesis and conceptual framework

The resulting conceptual framework separates three navigation concerns. First, perception provides directional range and visual context. Second, high-level navigation determines goals, routes, following states, and recovery actions. Third, an independent safety layer decides whether physical motion remains permissible. This structure is important because even a correct planned route may become unsafe when the environment changes. The framework therefore combines deliberative planning with continuous local safety reevaluation rather than treating path planning as a one-time decision.

3 RESEARCH METHODOLOGY

3.1 Research design

A design-and-build research methodology was used. Navigation behaviours were first explored in simulation, where geometric conditions, human movement, obstacle layout, and state transitions could be controlled without risking hardware damage. Physically integrated functions were then evaluated on the BUDDY prototype. Evidence was classified into three levels: implemented software, simulation-observed behaviour, and physically validated behaviour. This distinction prevents simulation results from being presented as real-world performance.

3.2 Data participants or study materials

The main study materials were the BUDDY Raspberry Pi 4 platform, four HC-SR04 ultrasonic sensors, four-wheel drive base, ROS 2 Jazzy software environment, and the navigation simulation. The 3D simulation represents a domestic environment with living, dining, and bedroom areas, walls, doorways, furniture obstacles, one principal human target, a second moving person, and the robot. No formal human-subject study was conducted for navigation performance in the current thesis version.

3.3 Experimental or study procedure

The physical evaluation followed the procedure already reported for the integrated BUDDY prototype [1]. Each of the four ultrasonic channels was exercised to verify directional observations. For dynamic obstacle intervention, BUDDY was commanded to move forward in a clear condition. An obstacle was then introduced into the front sensing region while the movement request remained active. The response of the safety supervisor and motor executor was observed. The obstacle was subsequently removed to determine whether the still-active movement request could be reconsidered.

The simulation procedure used repeated runtime observation of the robot in the furnished environment. The virtual human moved between rooms and tasks. The robot transitioned between wander, approach, follow, search, signal, and waiting states depending on target visibility and follow mode. A* replanning, obstacle

repulsion, path clearance, and PID-based distance control were exercised as the target and obstacles changed relative to the robot.

3.4 Measures and evaluation metrics

The current evaluation is primarily functional. Physical measures include successful acquisition of all four ultrasonic channels, successful interruption of unsafe forward movement, and movement resumption after obstacle removal. The simulation is evaluated through observable state correctness, collision avoidance, successful route generation, target-following behaviour, and target-loss recovery. No unsupported generalized navigation-accuracy percentage, mapping accuracy, collision rate, or localization error is reported.

Evidence category	Example	Claim level
Physical validation	Ultrasonic sensing; obstacle stop/resume	Strongest evidence in current thesis
Simulation validation	A* routing; follow/search state logic	Algorithmic behaviour demonstrated in simulation
Continuing work	2D occupancy-style planning; patrol; localization	Design direction, not yet physically validated

Table 3.1 Evidence classification used in this thesis

3.5 Data analysis

Results were analysed descriptively. Physical experiments were interpreted from observed system behaviour and logged software state rather than from large statistical datasets. Simulation analysis examined whether transitions and routes were consistent with the intended control logic. This approach is appropriate for a prototype-stage engineering thesis, but it limits generalisation and motivates future repeated-trial experiments.

3.6 Ethics privacy security and responsible AI

Navigation research creates physical safety and privacy considerations. The robot must not assume that a path is safe merely because a high-level planner requests motion. BUDDY therefore enforces a safety-supervision boundary. Camera-based navigation and future mapping should also avoid unnecessary storage of identifiable imagery. Any future deployment involving household maps or security patrol records should use local storage, access control, and explicit consent where people may be observed.

Generative AI was used as a drafting and language-assistance tool for this thesis. Technical statements were checked against project artefacts. AI-generated claims were not treated as experimental evidence, and unverified performance statistics were intentionally excluded.

3.7 Reproducibility and research artefacts

The principal reproducibility artefacts are the R26-SE-040 GitHub repository, the candidate's 'Yohan' development branch (commit 19a10e2c2d8a8ca18447ec0fd570ea75ce1954bd), `buddy_follow_me.py`, the integrated ROS 2 project branches, the group research paper, and the project website repository. The simulation encodes room geometry, navigation-cell size, obstacle clearance, PID gains, field-of-view angles, timeouts, and state transitions as explicit parameters. This makes the implementation logic inspectable and reproducible, although exact runtime trajectories vary because spawning, task selection, and behaviour timing use randomisation without a fixed seed.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirement	Description
R1	Detect obstacles around the mobile base using directional ultrasonic sensing.
R2	Prevent unsafe physical movement when an obstacle is detected or sensor information becomes stale.
R3	Plan collision-aware routes through an indoor environment in simulation.
R4	Follow a moving human while maintaining a practical distance and avoiding furniture.
R5	Recover when the target becomes occluded or temporarily leaves the field of view.
R6	Support future patrol, localization, mapping, and security-event behaviours without bypassing the safety layer.
R7	Operate within Raspberry Pi-class compute and low-cost sensor constraints.

Table 4.1 Navigation component requirements

4.2 Component architecture within the complete solution

The navigation architecture receives environmental observations and high-level movement intent, generates route or behaviour decisions, and then submits movement through the independent safety boundary. The safety supervisor can override or stop an active movement request when obstacle state changes. This separation is central to the overall BUDDY architecture [1].

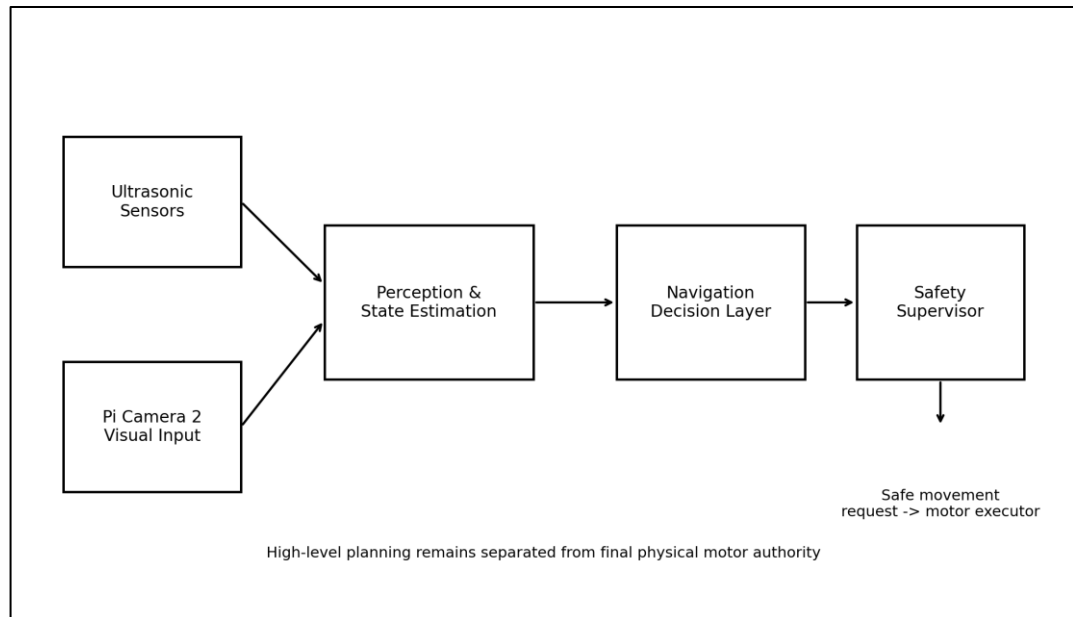


Figure 4.1 Navigation and safety architecture for the individual component

4.3 Design alternatives and decisions

A full SLAM stack with LiDAR would provide stronger metric localization and mapping, but it would significantly increase hardware cost and exceed the initial prototype scope. The selected approach therefore begins with directional ultrasonic sensing for reliable local range checks and uses simulation to develop higher-level planning logic. A monocular Raspberry Pi Camera 2 is retained for visual context and future semantic/navigation cues, but it is not treated as a direct depth sensor.

A* was selected for simulated route planning because the indoor environment can be represented as a grid and because the algorithm provides deterministic, interpretable route search with a well-established theoretical basis [4]. The simulation adds a clearance-aware cost map so that routes near walls and furniture become more expensive, reducing the tendency to skim obstacles.

PID-based distance control was selected for follow-me behaviour because it provides an understandable mechanism for regulating the difference between current and desired following distance. The design also clamps the integral term, rate-limits the visual heading rotation, and smooths acceleration to reduce abrupt simulated motion.

4.4 Detailed design

The simulated navigation grid uses fixed cells, blocked-cell inflation for walls and obstacles, diagonal and orthogonal A* neighbours, and a proximity cost that increases near blocked cells. A line-of-sight function evaluates whether a segment between robot and target is obstructed. When direct access is blocked, the system can search for a temporary peek position near obstacle corners.

Target following is managed by a state machine. The robot may wander when follow mode is inactive, approach an audio or last-seen cue, follow a visible target, search when visibility is lost, signal after search timeout, and wait for a renewed cue. This design makes the robot behaviour recoverable rather than assuming continuous visual contact.

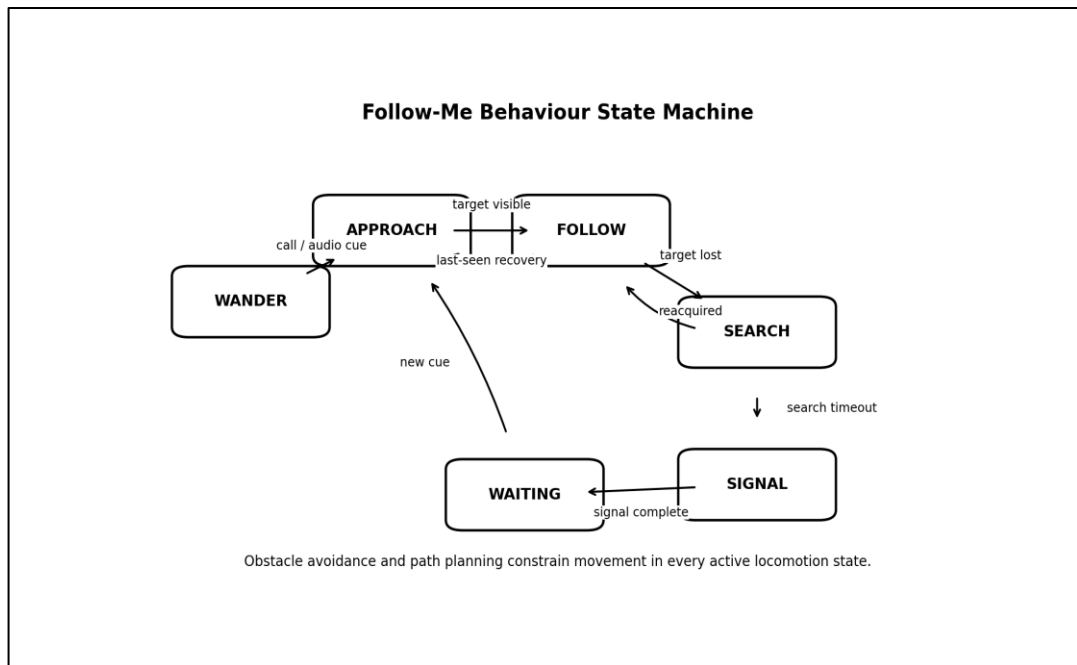


Figure 4.2 Follow-me behaviour state machine implemented in simulation

The simulation models an 80-degree primary field of view and a wider soft visibility region, checks occlusion against furniture, uses a grace period before declaring the target lost, and maintains last-seen information for recovery. A second human is included as a moving dynamic object so that avoidance behaviour is not restricted only to static furniture.

4.5 Standards and professional practice

The software follows modular ROS 2 engineering principles by separating sensing, decision, safety, and actuation responsibilities. Physical GPIO and motor control are kept behind hardware-oriented interfaces rather than embedded directly in high-level behaviour. The thesis also follows professional reporting practice by distinguishing configuration parameters from measured performance and by separating simulated, implemented, and physically validated results.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Technology	Role
Raspberry Pi 4B, 4 GB	Embedded computer for the physical robot
Ubuntu 64-bit + ROS 2 Jazzy	Robot middleware and runtime environment
Python	Navigation logic and simulation implementation
Ursina	3D visualisation engine for the 2D navigation simulation
heapq / custom A*	Grid-based path planning
4 x HC-SR04	Directional ultrasonic obstacle sensing
Raspberry Pi Camera Module 2 / IMX219	Monocular visual input for person/environment cues
TB6612FNG motor drivers	Physical drive control through the shared robot platform
Next.js / React / TypeScript	Project website contribution

Table 5.1 Development and implementation stack

5.2 Component implementation

The candidate's `Yohan` development branch in the R26-SE-040 repository contains `buddy_follow_me.py`, a self-contained 3D-visualised 2D navigation and follow-me simulation. The program creates a domestic floor layout, furniture obstacles, navigation grid, cost map, human actors, and a simplified dog-like visual robot representation. The implementation includes collision geometry, safe spawn logic, route planning, target visibility checks, obstacle-aware movement, camera visualization, and behaviour state management.

The A* planner maps world coordinates into navigation cells, searches eight-connected neighbours, applies a higher traversal cost near blocked cells, reconstructs a waypoint sequence, and provides the resulting route to the movement controller. Replanning is triggered as the target or current navigation goal changes.

The follow controller uses proportional, integral, and derivative terms to regulate distance to a heel position behind and slightly beside the human. The integral term is clamped, acceleration is smoothed, and visual heading rotation is rate-limited to approximately 200 degrees per second. These choices were introduced to reduce abrupt simulated motion compared with instant heading changes or direct pursuit.

Obstacle avoidance adds a local repulsive vector when the robot approaches walls, furniture, or the second moving person. The repulsive direction is blended with the planned navigation direction. A separate physical safety supervisor remains responsible for final movement authorization on the real robot; simulation avoidance is therefore a planning behaviour rather than the only safety mechanism.

5.3 Integration with the other components

The navigation component integrates with voice through high-level movement requests and with the visual component through target position and future person-following cues. It integrates with the safety subsystem through environmental observations and safety-approved movement topics. The project paper reports the implemented control flow as movement request -> safety supervisor -> safe movement -> motor executor [1].

The candidate also developed the BUDDY project website repository, where navigation is presented as a dedicated research capability and the team contribution is documented as navigation, autonomous patrol, security functionality, A* planning, dynamic replanning, remote-control requirements, patrol status, security alerts, and live-camera requirements. This website contribution is treated as supporting product documentation rather than as evidence of navigation performance.

5.4 Verification and software testing

Software-level verification was carried out at both group and component levels. The group research paper reports 13 relevant movement and safety tests passing during development [1]. The navigation simulation was repeatedly exercised to verify path generation, state transitions, target reacquisition, collision constraints, and stable following. Because no controlled statistical simulation campaign was recorded, the results are reported as functional verification rather than as a numerical success rate.

Test	Environment	Expected behaviour	Status
Four-direction ultrasonic availability	Physical robot	All four sensing positions produce usable observations	Passed
Dynamic obstacle insertion	Physical robot	Active forward movement is interrupted	Passed
Obstacle removal	Physical robot	Still-active forward request can resume after safety reevaluation	Passed
A* route generation	3D-visualised 2D simulation	Route is generated around blocked cells	Functionally observed
Target following	3D-visualised 2D simulation	Robot maintains following behaviour while target moves	Functionally observed
Target loss / search	3D-visualised 2D simulation	Robot transitions to recovery states rather than continuing blindly	Functionally observed
2D occupancy-style mapping	Design / ongoing	Combine range and visual cues into local planning grid	Pending physical validation

Table 5.2 Navigation verification summary

5.5 Deployment and operation

On the physical robot, ultrasonic sensors are triggered sequentially to reduce cross-interference. The group configuration uses caution and stop thresholds of approximately 0.45 m and 0.25 m respectively, while stale environmental information is treated as unsafe. These are configuration parameters, not calibrated accuracy claims [1]. Wheel-speed calibration identified approximately 0.2 normalized drive level as a practical straight-line value for the prototype, with higher levels required for turning [1].

6 RESULTS AND VALIDATION

6.1 Research results

The strongest research result is the demonstration that navigation intent and motor authorization can remain separate. During the dynamic obstacle experiment, the robot was already moving forward when an obstacle was introduced. The safety layer reevaluated the still-active request and stopped physical motion. After the obstacle was removed, the request was reconsidered and movement resumed [1]. This demonstrates continuous environmental reevaluation rather than a one-time obstacle check.

The second research result is the feasibility of a richer navigation strategy in simulation. The implemented model combines global route search, local obstacle influence, target visibility, state transitions, last-seen recovery, and PID-based following in one environment. This provides a structured basis for later transfer to the physical platform.

6.2 Product and technical results

Result	Evidence	Interpretation
Four ultrasonic directions available	Physical integration	Environmental sensing successfully integrated
Obstacle-triggered stop	Physical experiment	Safety supervisor can interrupt active motion
Movement resumes after obstacle removal	Physical experiment	Active request is continuously reevaluated
13 movement/safety software tests passed	Software test suite	Expected safety/control behaviour supported
A* and dynamic replanning implemented	Candidate simulation implementation	Advanced planning logic exists at simulation level

Follow/search/reacquisition behaviour implemented	Candidate simulation implementation	Human-following state logic exists at simulation level
Project navigation/security website section implemented	Website repository	Product communication and system requirements documented

Table 6.1 Summary of validated and implemented navigation results

6.3 User or stakeholder validation

No formal participant study was conducted for the navigation component. Stakeholder validation therefore comes from supervisor-directed scope evolution, team integration requirements, physical prototype observation, and the final system demonstration process. Future work should include repeated indoor route trials and structured observer evaluation of safety, following distance, recovery behaviour, and patrol usefulness.

6.4 Integration results

The physical navigation subsystem successfully interacts with the shared safety and motor-control architecture. The current system demonstrates that environmental sensing can override an ongoing movement request, which is particularly important when voice or future autonomous behaviours initiate motion. Complete integration of simulated follow-me planning, 2D mapping, and patrol behaviour into the physical runtime remains in progress.

6.5 Summary of findings

The component demonstrates a credible progression from local obstacle sensing to richer autonomous navigation. Physical evidence supports safety-mediated obstacle response. Simulation evidence supports A* route planning, state-based human following, and recovery logic. More advanced mapping, localization, patrol, and camera-assisted 2D planning should therefore be interpreted as the next stage of the research rather than as completed physical results.

7 DISCUSSION

7.1 Interpretation of findings

The physical obstacle experiment shows why navigation safety should not be delegated entirely to high-level planning. Even when the original forward request was acceptable, a later environmental change required a different physical outcome. The independent safety layer allowed this decision to be made without cancelling the high-level request itself. This architectural separation is useful for future autonomous patrol and follow-me behaviour because those behaviours can remain goal-oriented while lower-level safety continues to supervise movement.

The simulation demonstrates that useful companion behaviour requires more than shortest-path computation. A following robot must maintain distance, avoid walls and furniture, reason about visibility, recover after occlusion, and behave predictably when the target cannot be found. The implemented state machine captures these interaction-oriented requirements directly.

7.2 Comparison with literature baselines and existing solutions

A* provides the global graph-search foundation used by the simulation [4]. Occupancy-grid literature demonstrates how range observations can populate a raster world model [2], while probabilistic robotics provides the broader theoretical treatment of uncertainty in localization and mapping [3]. Dynamic Window and Vector Field Histogram approaches illustrate the importance of reactive collision avoidance during goal-directed motion [5], [6]. BUDDY combines related concepts at a simplified prototype scale while adding an explicit safety-supervision boundary that is central to the group architecture.

7.3 Research contribution

The individual research contribution is a low-cost navigation design that joins physical obstacle supervision with a richer simulation-based autonomy layer. The work provides an inspectable implementation of A* planning, clearance-aware grid costs, target-following state logic, PID distance regulation, dynamic replanning, visibility/occlusion reasoning, and recovery behaviour. It also establishes a

technically realistic path toward 2D occupancy-style navigation using low-cost range sensing and monocular vision without overstating the sensing capabilities of the camera.

7.4 Product contribution

At product level, the component provides the environmental-awareness and movement-intelligence foundation needed for autonomous patrol, follow-me interaction, security response, and mobile remote control. The candidate also implemented the BUDDY project website used to communicate research components, architecture, development milestones, team responsibilities, and navigation/security concepts.

7.5 Limitations and threats to validity

- Complete autonomous patrol has not yet been physically validated.
- The advanced follow-me and A* navigation behaviours are presently demonstrated in simulation rather than through a controlled physical trial campaign.
- No calibrated ultrasonic ranging-accuracy study has been completed; thresholds are configuration values rather than experimentally derived error bounds.
- Wheel odometry, IMU-based localization, and metric SLAM are not yet part of the validated navigation results.
- The Raspberry Pi Camera 2 is monocular and therefore does not directly provide true metric depth or a 3D point cloud.
- The proposed 2D occupancy-style mapping approach is continuing work and requires physical evaluation under different lighting, room layouts, and obstacle materials.
- Functional tests demonstrate feasibility but do not establish long-duration reliability or generalized collision-free performance.
- • The simulation has direct access to ground-truth obstacle geometry and exact simulated target coordinates. The physical robot does not yet possess equivalent global map knowledge or metric localization, so successful simulated A* and

follow-me behaviour should not be interpreted as equivalent autonomous performance on the physical platform.

7.6 Implications

The results suggest that useful safety and navigation behaviour can be developed incrementally on affordable hardware. A low-cost robot does not need to begin with a complete premium SLAM sensor suite to gain meaningful environmental awareness. However, advanced autonomy should be added progressively and validated separately from basic safety. This staged approach is particularly appropriate for educational and research companion robots where cost, inspectability, and modular experimentation are important.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The navigation component supports BUDDY's broader value proposition as an affordable, modular companion-robot platform. Potential users include educational institutions, robotics researchers, households interested in experimental companion robotics, and developers who need an accessible platform for testing human-robot interaction and indoor autonomy.

8.2 Market and competitive position

Commercial companion and mobile-robot platforms often combine sophisticated sensing with proprietary hardware and software. BUDDY is positioned differently: it prioritises replaceable components, open development tools, local processing, and progressive capability expansion. The navigation research contributes to this position by exploring what can be achieved with low-cost range sensors, a Raspberry Pi-class computer, and a standard monocular camera before adding more expensive sensors.

8.3 Business or sustainability model

A sustainable development model would use a modular base platform with optional capability packs for navigation, vision, voice, and security. This reduces replacement cost and allows educational users to upgrade selected modules. Open-source software components can support community development while hardware kits, integration services, training, or specialised deployments provide possible commercial pathways.

8.4 Scalability, intellectual property and compliance

Scalability depends on maintaining clear ROS 2 interfaces and keeping safety supervision independent of higher-level intelligence. Future commercial development would require electrical safety review, battery compliance, privacy policies for camera and map data, secure remote-control authentication, and more rigorous physical reliability testing. Intellectual-property decisions should distinguish original project software from third-party open-source frameworks and libraries.

8.5 Contribution to the Sustainable Development Goals

The component aligns most directly with SDG 9 through accessible innovation and engineering research. It also has potential relevance to SDG 3 when companion or assistive use cases are developed responsibly, and to SDG 4 through robotics education and practical STEM learning. These links describe potential contribution pathways rather than validated social impact.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the autonomous-navigation and obstacle-awareness contribution to BUDDY. The work combines physically validated ultrasonic safety intervention with a richer simulation environment for A* planning, human following, dynamic replanning, target-loss recovery, and collision-aware behaviour. The physical experiments confirm that environmental changes can interrupt an already-active movement request through the independent safety layer. The simulation demonstrates a practical architecture for extending the robot toward more capable indoor autonomy.

9.2 Achievement of objectives

- Safety-separated navigation architecture: achieved at physical system level.
- Four-direction ultrasonic integration: achieved and physically exercised.
- Dynamic obstacle interruption and resumption: achieved and physically demonstrated.
- A* route planning and collision-aware simulation: implemented and functionally observed.
- Human-following state machine and PID distance control: implemented in simulation.
- Camera-assisted 2D occupancy-style planning: exploratory sensor-fusion architecture proposed; implementation and physical validation pending.
- Patrol and localization: continuing work.

9.3 Individual research and product contributions

The candidate contributed the navigation research stream, including obstacle-aware movement design, advanced 3D simulation, A* path planning, dynamic replanning, follow-me behaviour, target recovery logic, and navigation/security system requirements. The candidate also developed the BUDDY project website used to communicate the system architecture, development state, research components, and team responsibilities.

9.4 Future directions

The next development stage is to transfer more of the simulated behaviour to the physical robot. Priority work includes full-body and person tracking integration, wheel odometry or encoder feedback, IMU integration, 2D occupancy-style local mapping, route execution, patrol-state management, repeated obstacle trials, and controlled follow-me experiments. If reliable metric localization is required, additional sensing such as depth, stereo, or LiDAR should be evaluated rather than overstating the capability of the monocular camera.

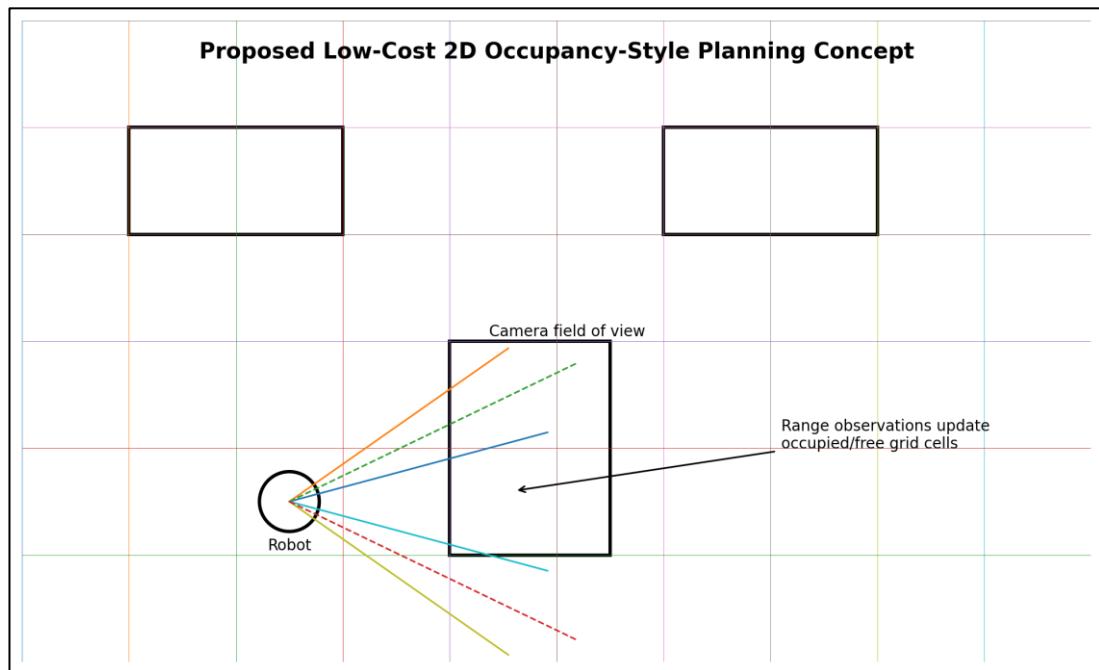


Figure 9.1 Proposed low-cost 2D occupancy-style planning concept for continuing work

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GLOSSARY

Term	Definition
A*	A heuristic graph-search algorithm used to find a low-cost path between a start and goal node.
Cost map	A grid in which traversal cost varies according to obstacle proximity or other constraints.
Dynamic replanning	Recomputing or updating a route while the target or environment changes.
Field of view	The angular region within which the simulated robot can observe a target.
Occupancy grid	A raster representation in which cells describe occupied, free, or unknown space.
Safety supervisor	Independent logic that decides whether a high-level movement request may be physically executed.
SLAM	Simultaneous localization and mapping; estimating robot pose while constructing a map.

APPENDICES

Appendix A Research instruments or extended results

A.1 Key simulation parameters

Parameter	Value in simulation	Purpose
Navigation cell size	7 simulation units	Grid resolution
Navigation clearance	13 units	Obstacle inflation
Primary field of view	80 degrees	Target visibility
Soft field of view	115 degrees	Graceful visibility loss
Lost-target grace period	0.45 s	Avoid immediate search transition
Search timeout	2.5 s	Transition from search to signal
Desired follow distance	Approx. 65 units	Human-following spacing
Maximum turn rate	Approx. 200 deg/s	Smooth visual rotation
PID gains	Kp 2.2, Ki 0.02, Kd 0.8	Follow-distance regulation

A.2 Current physical navigation configuration

Item	Current configuration / evidence
Ultrasonic layout	Front-left, front-right, rear-left, rear-right
Trigger spacing	Approximately 60 ms sequential separation
Caution threshold	Approximately 0.45 m
Stop threshold	Approximately 0.25 m

Stale-sensor handling	Environmental observations older than about 1.0 s treated as stale
Safety reevaluation	Approximately every 0.1 s during active movement
Motor watchdog	Approximately 0.05 s at executor layer
Straight movement calibration	Approximately 0.2 normalized drive level

Appendix B Academic integrity and AI use disclosure

Tool: ChatGPT. Purpose: assistance with thesis structure, language refinement, synthesis of existing project artefacts, literature organisation, and preparation/refinement of tables and figures. Verification: technical statements were checked against the R26-SE-040 source code, the BUDDY research paper, PP2 presentation material, and reported physical or software evidence. AI-generated suggestions were not treated as experimental results; the candidate remains responsible for the final thesis, citations, code ownership, and interpretation.

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